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Enhanced Grey Wolf Optimization for Efficient Transmission Power Optimization in Wireless Sensor Network

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Abstract: The Internet of Things (IoT) and Wireless Sensor Networks (WSNs) heavily rely on the lifetime of sensor nodes, which is inversely proportional to transmission power. Nodes with greater separation demand higher transmission power, while those closer together require less power. In practice, node placement varies significantly due to diverse terrain and contours, making power transmission configuration a critical and challenging issue in WSNs. This paper introduces an Enhanced Grey Wolf Optimization (EGWO) algorithm designed to optimize power transmission in WSN environments. Traditional Grey Wolf Optimization (GWO) employs a parameter that decreases linearly with iterations to regulate exploitation. In contrast, the proposed EGWO adopts a concave decline in the exploitation rate, allowing for more precise optimization in areas under exploration. The enhancement utilizes a cosine function that gradually decreases from 1 to 0, providing a smoother and more controlled transition. Experimental results demonstrate that EGWO outperforms other optimization algorithms. The proposed method achieves the lowest fitness value of -4.21, compared to 1.22 for standard GWO, -2.81 for PSO, and 2.86 for BESO, indicating its superiority in optimizing power transmission in WSNs.

Keywords: Energy Efficiency; Grey Wolf Optimization; Intra-Cluster Communication; Swarm Algorithms; Wireless Sensor Networks

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1. Introduction

The global industrial wireless sensor network market is experiencing significant growth, with projections estimating it will reach $16.1 billion by 2028, driven by a compound annual growth rate (CAGR) of 18.21%. This growth is attributed to the increasing adoption of industrial automation, advancements in wireless communication technologies, and the rising demand for real-time monitoring solutions. Industries are leveraging these networks to enhance operational efficiency, reduce costs, and improve predictive maintenance capabilities. The widespread deployment of wireless sensor networks is also fueled by their scalability and flexibility, making them a critical component in the evolution of Industry 4.0. [1].

Although the benefits of WSN integration within IoT ecosystems are vast, there are several challenges that hinder optimal performance. Key issues such as power consumption, data processing, and security vulnerabilities are major concerns affecting the reliability and longevity of these networks. Effectively addressing these challenges is critical for ensuring the long-term sustainability and success of WSN-enabled IoT solutions [2].

The convergence of WSNs and IoT technologies has unlocked numerous opportunities across various industries; however, addressing energy efficiency and security challenges is

crucial for their long-term viability. Since WSN nodes rely on limited power sources, efficient energy management is essential, particularly in scenarios where battery replacement is challenging. Researchers are investigating methods such as energy-efficient protocols and energy harvesting techniques to prolong operational life. Additionally, security remains a pressing issue due to the absence of standardized frameworks, prompting the adoption of lightweight encryption, blockchain solutions, and trust-based models. Advancements in these areas are vital to developing scalable, reliable, and energy-efficient IoT-driven WSN systems. [3].

In WSN, clustering is crucial for improving efficiency and extending network life. Clustering groups sensor nodes into clusters led by a Cluster Head (CH), which collects data from nodes and sends it to the base station, reducing communication and saving energy. Various algorithms optimize CH selection to maximize network lifetime by minimizing energy consumption and balancing load distribution. Notable algorithms include LEACH, which uses random CH rotation [4]; HEED, focusing on residual energy and neighbor density [5]; PEGASIS, forming a sensor chain to reduce energy use [6]; and DEEC [7] and TEEN [8], offering innovative CH selection strategies based on energy and thresholds.

In WSN, swarm algorithms significantly optimize network performance. Inspired by natural behaviors, these algorithms regulate node interactions within clusters, enhancing data collection and transmission while reducing energy consumption. For instance, Particle Swarm Optimization dynamically adjusts routes within clusters, improving communication efficiency and reducing CH load [9]. Ant Colony Optimization optimizes path selection by finding the shortest, most energy-efficient routes [10]. Genetic Algorithms [11] and Bee Colony Optimization [12] also improve resource allocation and management, thereby reducing energy use and increasing network life and efficiency [13].

This study addresses a research gap in WSN optimization, inspired by Silva Fre et al. [14] on energy optimization. Despite progress, there is potential for improvement in using swarm algorithms for transmission power optimization. While many strategies exist to save energy and extend node life in WSNs, few focus on swarm algorithms.

The main contribution of this study is the development of an Enhanced Grey Wolf Optimization (EGWO) using a non linear decreasing exploitation parameter that allows more precise exploration in a area. The proposed EGWO is further evaluated in four different topology with other three swarm optimization methods namely, standard GWO, BESO, and PSO. The performance of swarm algorithms is assessed using the Friis transmission power model, a standard for determining distance and signal strength in wireless communications. This model evaluates how each algorithm optimizes transmission power in WSNs to maximize energy efficiency while maintaining high communication quality.

The paper has four main sections: 'Introduction,' outlining WSN background and challenges; 'Methodology,' describing the technical framework; 'Results,' presenting experimental findings; and 'Conclusions,' evaluating the results and future research directions.

2. Methodology

This research is based on work by Silva Fre et.al [14] on optimal energy transmission using Particle Swarm Optimization. The proposed methodology is to compare other metaheuristic or swarm techniques to see which one yields the lowest transmission power, considering the limitations on transmit or receive capabilities. In this section, we will discuss the Friis model, swarm algorithm and experimental scenarios.

2.1. WSN Transmission Power

The Friis formula ,

$$P_R = \frac{A_r A_t}{d^2 \lambda^2} P_t \tag{1}$$

Formula for receiving power with effective cross-sectional area of antenna (1).

$$A_{\text{isotropik}} = \frac{\lambda^2}{4\pi} \tag{2}$$

Formula for isotropic antenna,

$$P_R = \left(\frac{\lambda}{4\pi d}\right)^2 P_t \tag{3}$$

Simplified formula for receiving power with an isotropic antenna. The antenna used is isotropic (an ideal antenna that radiates with equal power in all directions)(2), so its gain is 1 or does not need to be calculated.

The Friis formula is used as an objective function to find the optimal (lowest) transmission power value while meeting operational constraints (3). The objective function includes minimizing the transmitted power (P_t) while ensuring that the received power (P_R) is sufficient for reliable communication between nodes.

To ensure connectivity between nodes, a threshold (ρ_{th}) check of -60 dB is used (4).

$$A = \begin{cases} 1, & \text{if } P_R \geq \rho_{\text{th}} \\ 0, & \text{otherwise} \end{cases} \tag{4}$$

2.2. Optimization Algorithms

The following algorithms are used for the optimization enhancement and performance comparison.

2.2.1. Grey Wolf Optimizer

Grey Wolf Optimizer (GWO) is a metaheuristic optimization algorithm inspired by the social hierarchy and hunting behavior of grey wolves (*Canis lupus*). In GWO, the population is divided into four hierarchical levels:

- Alpha (α): The best solution, leading the search.
- Beta (β): The second-best solution, assisting α in leadership.
- Delta (δ): The third-best solution, subordinate to α and β.
- Omega (ω): The remaining wolves, following α, β, and δ.

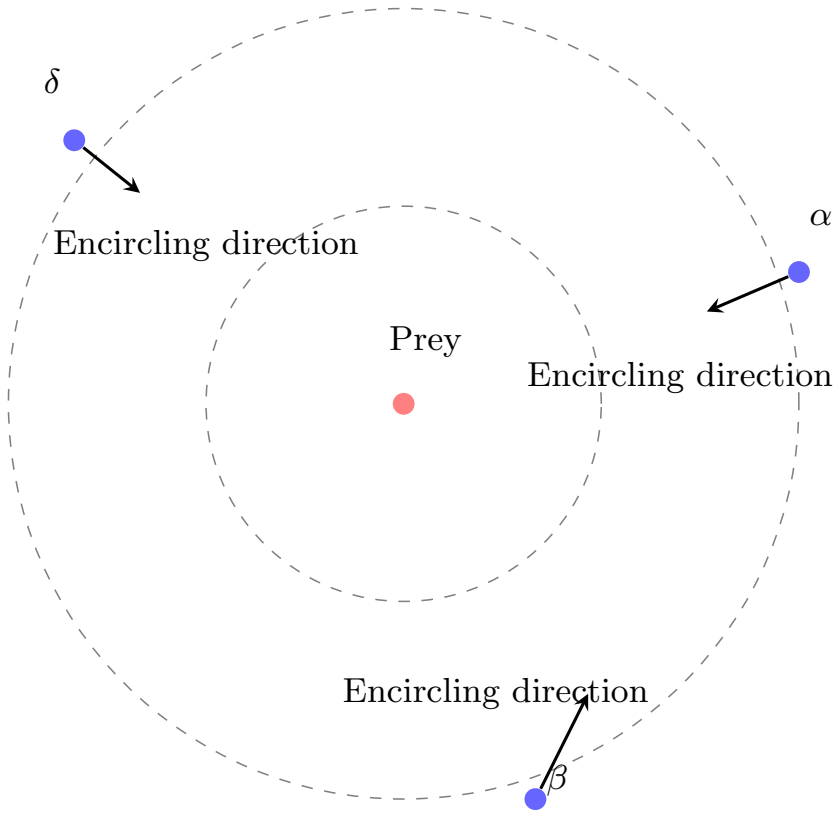


Figure 1. Grey Wolf Optimization

The optimization process in GWO mimics the hunting strategy of grey wolves as shown in Fig. 1, including encircling prey, searching for prey, and attacking prey. This mechanism is mathematically modeled as follows:

Encircling Prey

The encircling behavior of grey wolves is described by:

$$\mathbf{X}(t+1) = \mathbf{X}_p(t) - \mathbf{A} \cdot |\mathbf{C} \cdot \mathbf{X}_p(t) - \mathbf{X}(t)|, \tag{5}$$

where $\mathbf{X}(t)$ and $\mathbf{X}_p(t)$ denote the vector position of a wolf and a prey at iteration t, respectively, The exploitation rate is controlled by coefficient A defined as follows :

$$\mathbf{A} = 2 \cdot \mathbf{a} \cdot \mathbf{r}_1 - \mathbf{a}. \tag{6}$$

This coefficient governs the wolves to concentrate in a certain area. Whereas, the exploration coefficient C is defined as :

$$\mathbf{C} = 2 \cdot \mathbf{r}_2. \tag{7}$$

This coefficient governs the random exploration vector $\mathbf{r}_1, \mathbf{r}_2 \in [0, 1]$. The standard GWO uses a linearly decreasing parameter $\mathbf{a}$ from 2 to 0 over iterations as shown in Fig. 2.

Proposed Enhanced GWO with Non-Linear Decreasing Coefficient

This paper introduces an Enhanced GWO (EGWO) by introducing a decreasing coefficient with a non linear function namely cosine function. As shown in Fig. 2, the value of $\mathbf{a}$ at iteration t is given by :

$$\mathbf{a}(\mathbf{t}) = 2 \times \frac{T - t}{T}. \tag{8}$$

where T denotes the maximum number of iterations. Cosine is a concave function from 0 to $\frac{\pi}{2}$ that provides a slower decreasing rate than linear one. The proposed decreasing coefficient $\mathbf{a_c}(\mathbf{t})$ is given by :

$$\mathbf{a_c(t)} = 2 \times \cos \frac{\pi t}{2T} \tag{9}$$

Lemma 1. *For $t \in [0, T]$, the inequality $a(t) < a_c(t)$ holds*

Proof. To prove the inequality $a(t) < a_c(t)$, consider the difference between $a_c(t)$ and $a(t)$, defined as:

$$\Delta(t) = a_c(t) - a(t). \tag{10}$$

Substituting the expressions for $a(t)$ and $a_c(t)$, we get:

$$\Delta(t) = 2 \cdot \cos\left(\frac{\pi t}{2T}\right) - 2 \cdot \frac{T-t}{T}. \tag{11}$$

Factoring out 2, this simplifies to:

$$\Delta(t) = 2\left[\cos\left(\frac{\pi t}{2T}\right) - \frac{T-t}{T}\right]. \tag{12}$$

Let $f(t) = \cos\left(\frac{\pi t}{2T}\right) - \frac{T-t}{T}$. To analyze the behavior of $f(t)$, we evaluate it at the boundaries of the interval and examine its derivative.

At $t = 0$:

$$f(0) = \cos(0) - \frac{T-0}{T} = 1 - 1 = 0. \tag{13}$$

At $t = T$:

$$f(T) = \cos\left(\frac{\pi T}{2T}\right) - \frac{T-T}{T} = \cos\left(\frac{\pi}{2}\right) - 0 = 0. \tag{14}$$

To analyze the monotonicity of $f(t)$, compute its derivative:

$$f'(t) = \frac{d}{dt}\left[\cos\left(\frac{\pi t}{2T}\right) - \frac{T-t}{T}\right]. \tag{15}$$

The derivative is:

$$f'(t) = -\frac{\pi}{2T}\sin\left(\frac{\pi t}{2T}\right) + \frac{1}{T}. \tag{16}$$

Factoring out $\frac{1}{T}$, we can rewrite:

$$f'(t) = \frac{1}{T}\left[1 - \frac{\pi}{2}\sin\left(\frac{\pi t}{2T}\right)\right]. \tag{17}$$

Since $\sin\left(\frac{\pi t}{2T}\right) \in [0, 1]$ for $t \in [0, T]$, the term:

$$1 - \frac{\pi}{2}\sin\left(\frac{\pi t}{2T}\right) > 0,$$

because $\frac{\pi}{2} \approx 1.57 > 1$. Therefore, $f'(t) > 0$ for all $t \in [0, T]$, indicating that $f(t)$ is strictly increasing on this interval.

Since $f(t) = 0$ at both $t = 0$ and $t = T$, and $f(t)$ is strictly increasing on $[0, T]$, it follows that:

$$f(t) > 0 \quad \forall t \in (0, T).$$

This implies:

$$\cos\left(\frac{\pi t}{2T}\right) > \frac{T-t}{T} \quad \forall t \in [0, T].$$

Equivalently:

$$a(t) < a_c(t), \quad \forall t \in [0, T].$$

This completes the proof. □

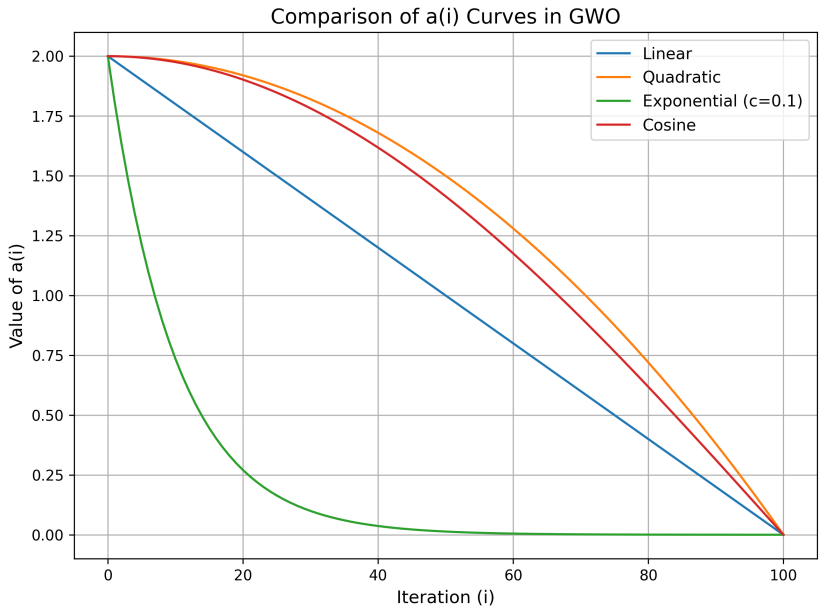


Figure 2. Comparison of a(i) Curves in GWO

Hunting Behavior

The hunting strategy is modeled based on the positions of α, β, and δ, as they guide the ω wolves:

$$\mathbf{X}(t+1) = \frac{\mathbf{X}_1 + \mathbf{X}_2 + \mathbf{X}_3}{3}, \tag{18}$$

where:

$$\mathbf{X}_1 = \mathbf{X}_\alpha - \mathbf{A}_1 \cdot |\mathbf{C}_1 \cdot \mathbf{X}_\alpha - \mathbf{X}(t)|, \tag{19}$$

$$\mathbf{X}_2 = \mathbf{X}_\beta - \mathbf{A}_2 \cdot |\mathbf{C}_2 \cdot \mathbf{X}_\beta - \mathbf{X}(t)|, \tag{20}$$

$$\mathbf{X}_3 = \mathbf{X}_\delta - \mathbf{A}_3 \cdot |\mathbf{C}_3 \cdot \mathbf{X}_\delta - \mathbf{X}(t)|. \tag{21}$$

Here, $\mathbf{X}_\alpha, \mathbf{X}_\beta, \mathbf{X}_\delta$ are the positions of the top three wolves, while $\mathbf{A}_1, \mathbf{A}_2, \mathbf{A}_3$ and $\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3$ are defined similarly to $\mathbf{A}$ and $\mathbf{C}$.

Convergence and Termination

The algorithm iteratively updates the positions of wolves until a termination criterion is satisfied, such as achieving the maximum number of iterations $T_{\max}$ or reaching a predefined fitness threshold.

Application in WSNs

In this study, GWO is applied to optimize QoS in WSNs for IoT applications. Key objectives include:

- Energy Efficiency: Minimizing energy consumption by optimizing cluster head selection and communication paths.
- Delay Reduction: Reducing end-to-end latency for time-sensitive IoT applications.

- Network Throughput: Enhancing packet delivery ratio and overall network performance.

The fitness function for GWO in this context is defined as:

$$f = \alpha \cdot \frac{1}{E} + \beta \cdot \frac{1}{D} + \gamma \cdot T, \tag{22}$$

where:
- E: Total energy consumption,
- D: Average delay in data transmission,
- T: Total network throughput,
- α, β, γ: Weighting factors balancing the objectives.

This approach enables dynamic resource allocation and efficient network management, addressing the unique challenges of WSNs in IoT applications [15].

2.2.2. Particle Swarm Optimization

Particle Swarm Optimization (PSO) is an optimization technique inspired by the social behavior of birds or fish. In PSO, each particle, representing a potential solution, explores the search space based on its velocity and adjusts its position according to its own experience and that of its peers. This study uses PSO for inter-cluster power allocation, considering Quality-of-Service (QoS) factors and energy consumption. Additionally, Adaptive Particle Swarm Optimization (APSO) is introduced for intra-cluster resource allocation, optimizing node distance and energy load [16].

The velocity and position updates of each particle are governed by the following equations:

$$v_i^{(t+1)} = w \cdot v_i^{(t)} + c_1 \cdot r_1 \cdot \left(p_i^{\text{best}} - x_i^{(t)}\right) + c_2 \cdot r_2 \cdot \left(g^{\text{best}} - x_i^{(t)}\right), \tag{23}$$

$$x_i^{(t+1)} = x_i^{(t)} + v_i^{(t+1)}, \tag{24}$$

where:
- $v_i^{(t)}$: velocity of particle i at iteration t,
- $x_i^{(t)}$: position of particle i at iteration t,
- p_i^{best}: personal best position of particle i,
- g^{best}: global best position in the swarm,
- w: inertia weight, controlling the balance between exploration and exploitation,
- c_1, c_2: cognitive and social coefficients, respectively,
- r_1, r_2: random numbers uniformly distributed in $[0, 1]$.

By adjusting the parameters w, c_1, and c_2, the PSO algorithm adapts its search behavior to balance exploration and exploitation.

In this study, APSO modifies the inertia weight w dynamically to improve convergence. The adaptive inertia weight is defined as:

$$w = w_{\max} - \frac{(w_{\max} - w_{\min}) \cdot t}{T_{\max}}, \tag{25}$$

where $w_{\max}$ and $w_{\min}$ are the maximum and minimum inertia weights, t is the current iteration, and $T_{\max}$ is the maximum number of iterations.

2.2.3. Bi-directional Evolutionary Structural Optimization

Bi-directional Evolutionary Structural Optimization (BESO) is a computational optimization method designed to achieve optimal structural designs by iteratively adding or removing materials based on performance criteria. Unlike traditional methods, BESO optimizes both material distribution and structural topology, making it particularly effective

for lightweight design and multi-scale problems.

Mathematical Formulation

The optimization process in BESO aims to minimize an objective function, such as compliance or weight, while satisfying design constraints. The design domain is discretized into finite elements, where each element is assigned a density variable $\rho_e \in [0, 1]$, representing the presence ($\rho_e = 1$) or absence ($\rho_e = 0$) of material.

1. Objective Function: The objective function for minimizing compliance is defined as:

$$C = \mathbf{U}^T \mathbf{K} \mathbf{U}, \tag{26}$$

where:

- C: Structural compliance,
- $\mathbf{U}$: Global displacement vector,
- $\mathbf{K}$: Global stiffness matrix.

2. Volume Constraint: The optimization problem must satisfy a volume constraint:

$$V = \sum_{e=1}^{N_e} \rho_e V_e \leq V_{\max}, \tag{27}$$

where:

- V: Total material volume,
- $V_{\max}$: Maximum allowable material volume,
- N_e: Total number of finite elements,
- V_e: Volume of element e.

3. Sensitivity Analysis: The sensitivity of the objective function with respect to the design variable ρ_e is calculated as:

$$\frac{\partial C}{\partial \rho_e} = -\mathbf{U}_e^T \mathbf{K}_e \mathbf{U}_e, \tag{28}$$

where $\mathbf{U}_e$ and $\mathbf{K}_e$ are the displacement vector and stiffness matrix of element e, respectively.

Convergence and Termination

The algorithm iterates until a termination criterion is met, such as reaching a predefined number of iterations $T_{\max}$ or when the change in the objective function ΔC between consecutive iterations falls below a tolerance threshold:

$$\Delta C = \frac{|C^{(t+1)} - C^{(t)}|}{C^{(t)}} \leq \epsilon, \tag{29}$$

where ϵ is the convergence tolerance.

4. Evolutionary Update Rule: The material distribution is updated iteratively using:

$$\rho_e^{(t+1)} = \begin{cases} 1, & \text{if } \frac{\partial C}{\partial \rho_e} \leq \eta, \\ 0, & \text{if } \frac{\partial C}{\partial \rho_e} > \eta, \end{cases} \tag{30}$$

where η is a threshold determined by the volume constraint and evolutionary ratio. 5.

Bi-directional Adjustment: BESO allows both addition and removal of materials, enabling a balanced search for optimal topology:

$$\rho_e^{(t+1)} = \rho_e^{(t)} + \Delta\rho, \tag{31}$$

where $\Delta\rho$ is the material adjustment increment.

Application in Multi-Scale Optimization

In this study, BESO is extended to simultaneous multi-scale topology optimization, addressing both cellular and composite materials. The fitness function incorporates separate and unified volume constraints, enabling the optimization of hierarchical structures. The process is defined as:

$$f(\mathbf{æ}) = \alpha \cdot C(\mathbf{æ}) + \beta \cdot P(\mathbf{æ}), \tag{32}$$

where:

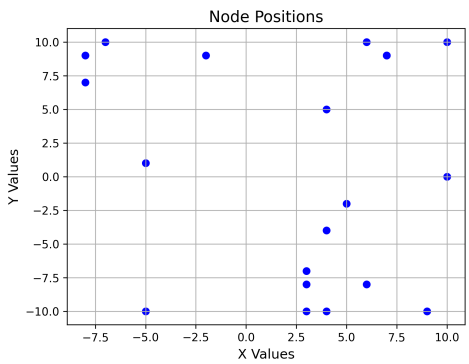
- $C(\mathbf{æ})$: Compliance of the structure,
- $P(\mathbf{æ})$: Penalization term for multi-scale connectivity,
- α, β: Weighting factors balancing the objectives.

Performance in Structural Design

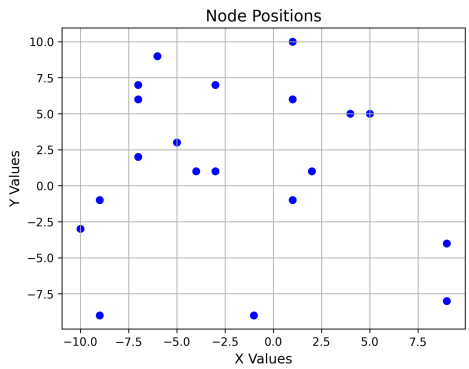
Kazakis et al. demonstrated the effectiveness of BESO for multi-scale optimization using a simplified MATLAB framework. The method supports cellular and composite materials while handling both separate and single volume constraints, showing significant improvements in structural performance and material efficiency [17].

2.3. Experiment scenario

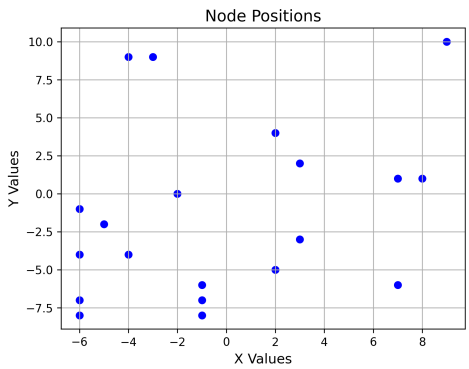
To find out the best optmization algorithm in general, four scenarios were made, with different node placements in each cluster. Figures 3a, 3b, 3c, and 3d display the positions of nodes represented by blue dots within a 20m x 20m area. The X and Y values, ranging from -10 to 10 meters, denote the relative locations of the nodes within this space. On a dimension scaled to 20m x 20m, the point at coordinates (-5, 5) on the original scale of -10 to 10 would correspond to coordinates (5.0m, 15.0m) on the new scale.



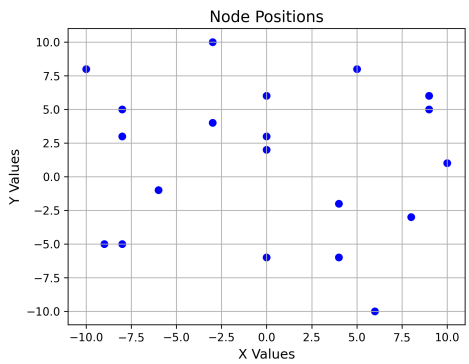
(a) Scenario 1



(b) Scenario 2



(c) Scenario 3



(d) Scenario 4

Figure 3. The Topology Utilized in the Experiments

3. Result and Discussion

We conducted experiments across four scenarios using the parameters listed in Table 1. The initial experiment aimed to demonstrate that the simplistic method (transmission power of -10 dBm or 0.0001 Watt, as suggested by [14]) can be optimized using swarm algorithms. As shown in Fig. 4, applying a standard swarm optimization method such as PSO results in lower power requirements compared to the simplistic method for all nodes. Building on these findings, we then focus on comparing the performance of various swarm optimization algorithms.

Table 1. Parameter, Value, Unit.

Parameter	Value [Unit]
Number of Sensor Nodes	20
Area for Random Sensors Distribution (LxL)	20 [m] x 20 [m]
Sensors Transmission Frequency (f)	915 [MHz]
Sensors Transmission Power Range (xMin; xMax)	[-30, 0] [dBm]
Sensor Sensitivity (ρ_{th})	-60 [dBm]

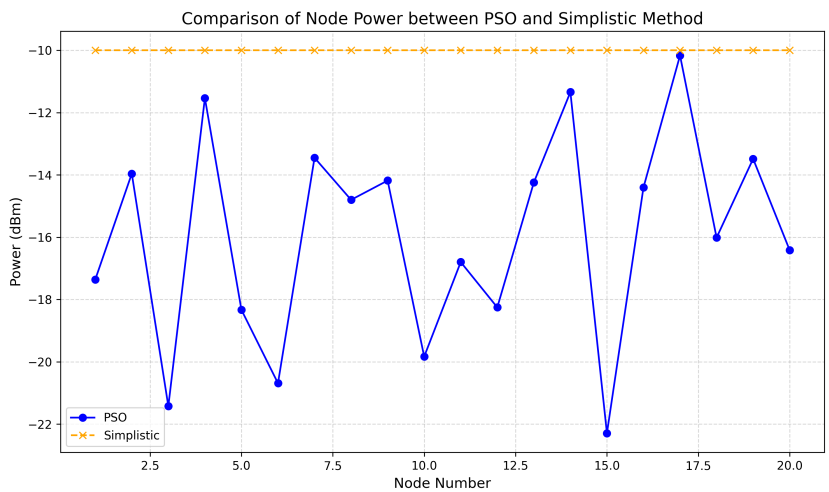


Figure 4. Power Transmission Simplistic vs PSO

Fig. 5 presents the average minimum fitness values achieved by all swarm algorithms across all scenarios. Notably, some algorithms, such as Simulated Annealing (SA) [18] and the Tuna Swarm Algorithm (TSA) [19], do not exhibit any fitness values, indicating that these algorithms failed to converge to any solution. Other algorithms, including Chicken Swarm Optimization (CSO) [20], Salp Swarm Algorithm (SSA) [21], Whale Optimization Algorithm (WOA) [22], and BESO, display positive fitness values, signifying convergence to suboptimal solutions. In contrast, algorithms like Particle Swarm Optimization (PSO), Genetic Algorithm (GA) [5,23], and Grey Wolf Optimization (GWO) achieve negative fitness values, indicating that these methods are very close to optimal solutions. Consequently, we focus exclusively on these three swarm algorithms for the remaining experiments.

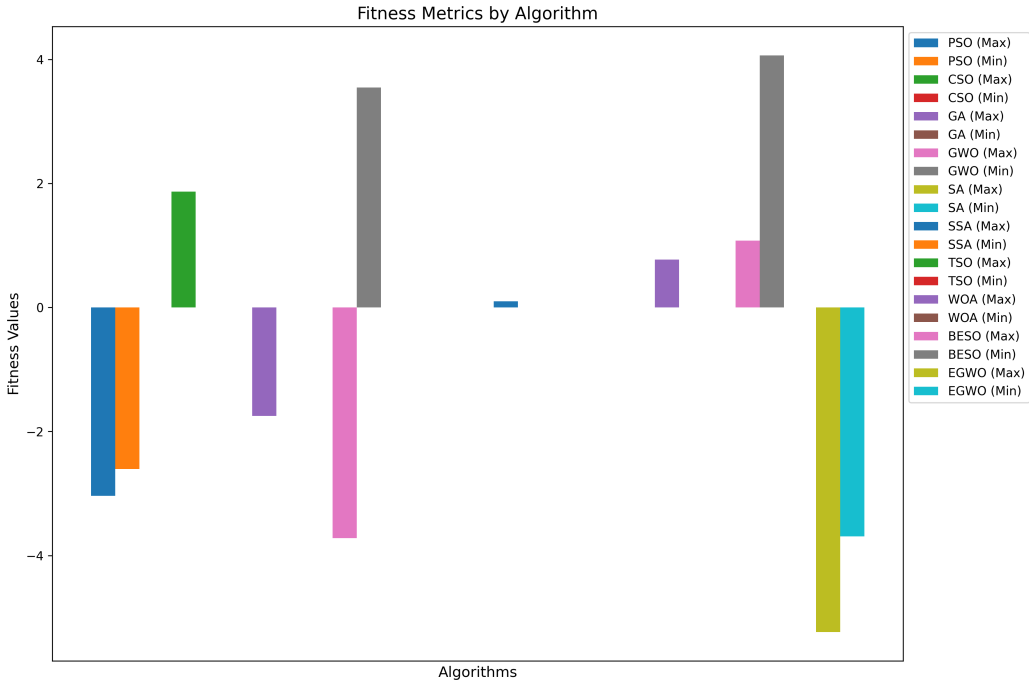


Figure 5. Fitness by Algorithm

If infinite values are observed in the Best Fitness test (Fig. 5) or NaN values appear in the Execution Time test (Table 2), only algorithms with finite values are included in the analysis. Based on the filtered results in Table 3, only EGWO, GWO, BESO, and PSO are considered for both Execution Time and Best Fitness metrics. Among these, GWO achieves the smallest Best Fitness value (-3.717954) with an execution time of 14.824434 seconds.

Table 2. Execution Time of each algorithm

Algorithm	Minimum	Maximum	Mean
BESO	12.511379	20.312532	14.615489
PSO	13.303567	33.346832	20.209724
GWO	14.824434	21.313970	16.637913
EGWO	20.05517125	21.6570003	20.65879297

Table 3 summarizes the swarm comparison in terms of fitness values. These statistics represent the results of the experimental comparison across all scenarios, where each scenario is repeated 10 times. GWO achieves the lowest minimum fitness value of -3.717, followed by PSO with -3.037. However, for other statistics, the lowest maximum and mean fitness values are achieved by PSO, with -2.605 and -2.81, respectively. Further details can be observed in Fig. 6, which illustrates the power transmission to each node for Scenario 1.

Table 3. Best Fitness of each algorithm

Algorithm	Minimum	Maximum	Mean
BESO	1.075610	4.064182	2.680685
PSO	-3.037325	-2.605397	-2.813640
GWO	-3.717954	3.547885	1.221277
EGWO	-5.235080	-3.689201	-4.212916

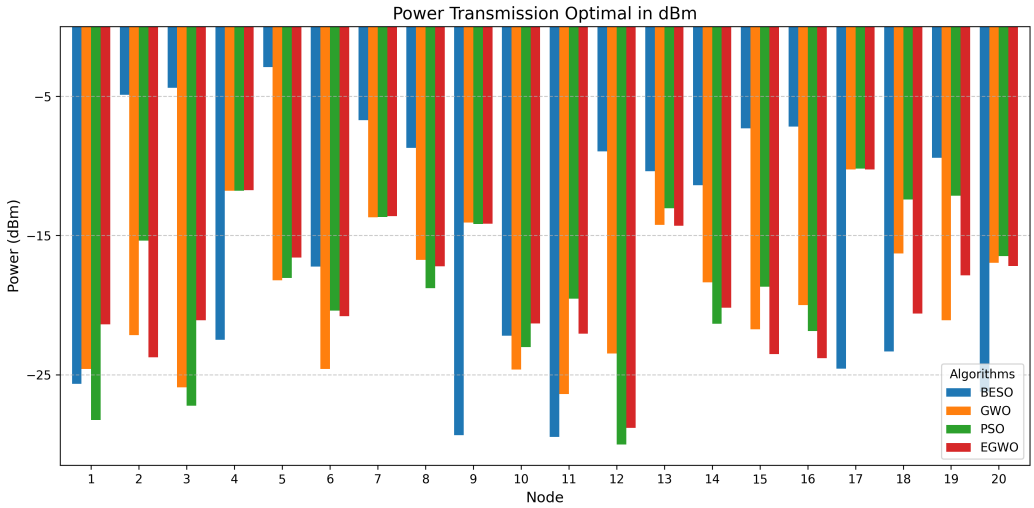


Figure 6. Power Transmission Values for Each Node in Scenario 1

From Fig. 6, it is evident that the three algorithms—GWO, PSO, and BESO—produce different power transmission values for each node, ranging from approximately -5 dBm to -30 dBm. For most nodes, the Grey Wolf Optimizer (GWO) achieves the lowest power transmission values (represented by the green bars), indicating that it frequently finds more optimal (lower) power solutions compared to PSO and BESO. BESO and PSO also perform well, with PSO often achieving results close to GWO. However, GWO demonstrates a clear advantage in achieving lower power transmission for most nodes, as seen in Fig. 7.

In Scenario 1, GWO achieves the lowest average power transmission, followed by PSO and BESO, indicating GWO's effectiveness in reducing power for this scenario. Similarly, in Scenario 2, GWO again shows the lowest average power transmission, with a smaller difference compared to PSO, while BESO continues to have the highest values. In Scenario 3, GWO maintains its lead with the lowest average power transmission, demonstrating consistent performance. However, PSO slightly underperforms compared to BESO in this scenario. In Scenario 4, BESO shows significant improvement, with an average power transmission nearly matching PSO and surpassing GWO, which records the highest power transmission (the least efficient) in this scenario. This analysis indicates GWO's consistent performance in most scenarios, except for Scenario 4, where BESO demonstrates notable improvement.

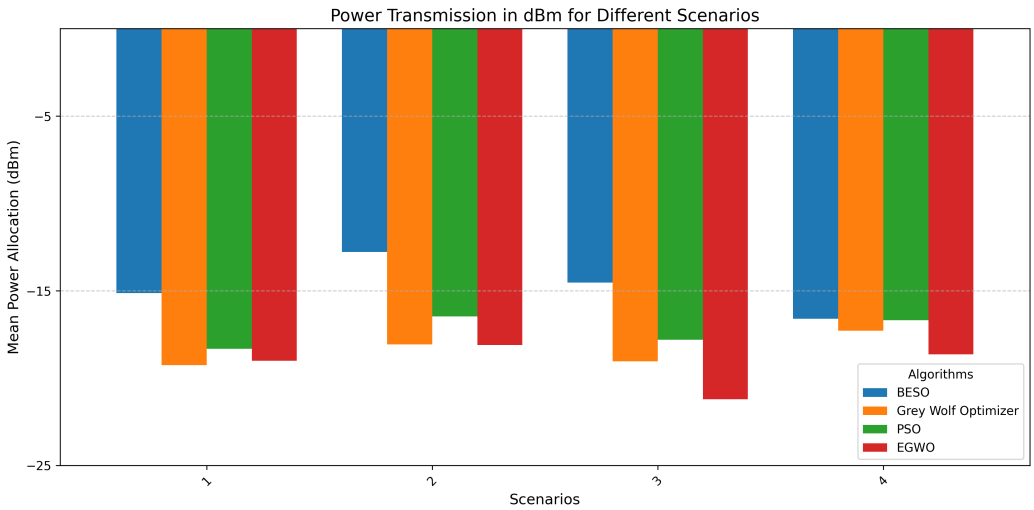


Figure 7. Power Transmission in 4 Scenarios

Table 4 shows the average value of optimal power transmission in dBm for four different scenarios, using four optimization algorithms: BESO, PSO, GWO, and EGWO. In Scenario 1, GWO achieves the lowest average power transmission with −19.256 dBm, followed by EGWO with −19.011 dBm, PSO with −18.323 dBm, and BESO with −15.139 dBm. This trend continues in Scenario 2, where GWO achieves −18.075 dBm, followed closely by EGWO with −18.113 dBm, PSO with −16.477 dBm, and BESO with −12.794 dBm.

In Scenario 3, EGWO outperforms all other algorithms, achieving the lowest power transmission of −21.216 dBm, followed by GWO with −19.042 dBm, PSO with −17.809 dBm, and BESO with −14.536 dBm. Similarly, in Scenario 4, EGWO again achieves the least power transmission with −18.652 dBm, while GWO, PSO, and BESO achieve −17.295 dBm, −16.693 dBm, and −16.608 dBm, respectively. These results emphasize the consistent performance of GWO and EGWO in minimizing power transmission, with EGWO demonstrating significant superior performance in Scenarios 3 and 4.

Table 4. Power transmission in dBm

Algorithm	Scenario			
	1	2	3	4
BESO	-15.139	-12.794	-14.536	-16.608
PSO	-18.323	-16.477	-17.809	-16.693
GWO	-19.256	-18.075	-19.042	-17.295
EGWO	-19.011	-18.113	-21.216	-18.652

Table 5 shows the average values of optimal power transmission converted from dBm to microWatts (μWatt) for four algorithms across four different scenarios. Similar to Table 4, GWO consistently achieves the lowest power transmission values in μWatt for most scenarios. For Scenario 1, GWO achieves an average power transmission of 11.8672 μWatt, followed closely by EGWO with 12.5570 μWatt, PSO with 14.7099 μWatt, and BESO with the highest value of 30.6239 μWatt.

In Scenario 2, GWO remains the most efficient algorithm with a transmission of 15.5750 μWatt, slightly better than EGWO with 15.4412 μWatt. PSO and BESO follow with 22.5040 μWatt and 52.5435 μWatt, respectively. Similarly, in Scenario 3, EGWO surpasses GWO with the lowest power transmission of 7.5565 μWatt, compared to GWO's 12.4670 μWatt. PSO and BESO record values of 16.5590 μWatt and 35.1882 μWatt, respectively.

In Scenario 4, EGWO again achieves a low power transmission of 13.6367 μWatt, while GWO, PSO, and BESO record values of 18.6421 μWatt, 21.4119 μWatt, and 21.8341 μWatt, respectively. These results confirm that GWO and EGWO consistently outperform PSO and BESO, with EGWO showing slightly better performance in Scenarios 3 and 4.

Table 5. Power transmission in μWatt

Algorithm	Scenario			
	1	2	3	4
BESO	30.6239	52.5435	35.1882	21.8341
PSO	14.7099	22.5040	16.5590	21.4119
GWO	11.8672	15.5750	12.4670	18.6421
EGWO	12.5570	15.4412	7.5565	13.6367

4. Conclusions

This study addresses a problem in WSN on energy optimization. While progress has been made in using swarm algorithms for transmission power optimization, there is still significant potential for improvement. Although many strategies exist to save energy

and extend node lifespan in WSNs, few focus specifically on swarm algorithms. This research introduced the Enhanced Grey Wolf Optimization (EGWO) algorithm, which utilizes a non-linear decreasing exploitation parameter to enable more precise exploration in a given area. The EGWO algorithm was evaluated against three other swarm optimization methods—standard GWO, BESO, and PSO—across four different topologies using the Friis transmission power model. This model, a standard for determining distance and signal strength in wireless communications, was used to assess how each algorithm optimizes transmission power in WSNs to maximize energy efficiency while maintaining high communication quality. Results demonstrate that EGWO averagely outperforms the other methods, achieving the best optimization results across all scenarios. For example, in Scenario 3, EGWO achieved a fitness value of -21.216, improving by 2.174 compared to GWO (-19.042) and by 3.407 compared to PSO (-17.809). These enhancements, including the improved balance between exploration and exploitation, make EGWO a highly effective and robust algorithm for power transmission optimization in WSNs.

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